

Signal Generator and Oscilloscope Experiments

Experiment 1: Series Resistor Circuit (Circuit A)

Given:

Peak-to-Peak Voltage (V_{pp}) = 5 V

Frequency (f) = 100 Hz

$R_1 = 5.1 \text{ k}\Omega$

$R_2 = 3.3 \text{ k}\Omega$

Step 1: Total Resistance

$R_T = R_1 + R_2 = 5.1\text{k} + 3.3\text{k} = 8.4 \text{ k}\Omega$

Step 2: Peak Voltage

$V_p = V_{pp} / 2 = 5 / 2 = 2.5 \text{ V}$

Step 3: Circuit Current using Ohm's Law

$I = V_p / R_T = 2.5 / 8400 = 0.000298 \text{ A} \approx 0.298 \text{ mA}$

Step 4: Voltage across each resistor

$V_1 = I \times R_1 = 0.298 \text{ mA} \times 5.1 \text{ k}\Omega \approx 1.52 \text{ V}$

$V_2 = I \times R_2 = 0.298 \text{ mA} \times 3.3 \text{ k}\Omega \approx 0.99 \text{ V}$

Step 5: Time Period

$T = 1 / f = 1 / 100 = 0.01 \text{ s} = 10 \text{ ms}$

Experiment 2: Series Resistor Circuit (Circuit B)

Given:

Peak-to-Peak Voltage (V_{pp}) = 7 V

Frequency (f) = 200 Hz

$R_1 = 4.7 \text{ k}\Omega$

$$R2 = 3.3 \text{ k}\Omega$$

$$R3 = 5.1 \text{ k}\Omega$$

Step 1: Total Resistance

$$R_T = R1 + R2 + R3 = 4.7\text{k} + 3.3\text{k} + 5.1\text{k} = 13.1 \text{ k}\Omega$$

Step 2: Peak Voltage

$$V_p = V_{pp} / 2 = 7 / 2 = 3.5 \text{ V}$$

Step 3: Circuit Current

$$I = V_p / R_T = 3.5 / 13100 = 0.000267 \text{ A} \approx 0.267 \text{ mA}$$

Step 4: Voltage across resistors

$$V1 = I \times R1 = 0.267 \text{ mA} \times 4.7 \text{ k}\Omega \approx 1.25 \text{ V}$$

$$V2 = I \times R2 = 0.267 \text{ mA} \times 3.3 \text{ k}\Omega \approx 0.88 \text{ V}$$

$$V3 = I \times R3 = 0.267 \text{ mA} \times 5.1 \text{ k}\Omega \approx 1.36 \text{ V}$$

Step 5: Time Period

$$T = 1 / f = 1 / 200 = 0.005 \text{ s} = 5 \text{ ms}$$